



CS & IT ENGINEERING



THEORY OF COMPUTATION

Regular Language

Lecture No. - 01



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Recap of Previous Lecture



Topic

?????

Regular Expression

Properties of Regular Expression

Finite Automata $\Leftrightarrow$ Regular Expression



Topics to be Covered



Topic

Regular Language Detection

Topic

Pumping Lemma



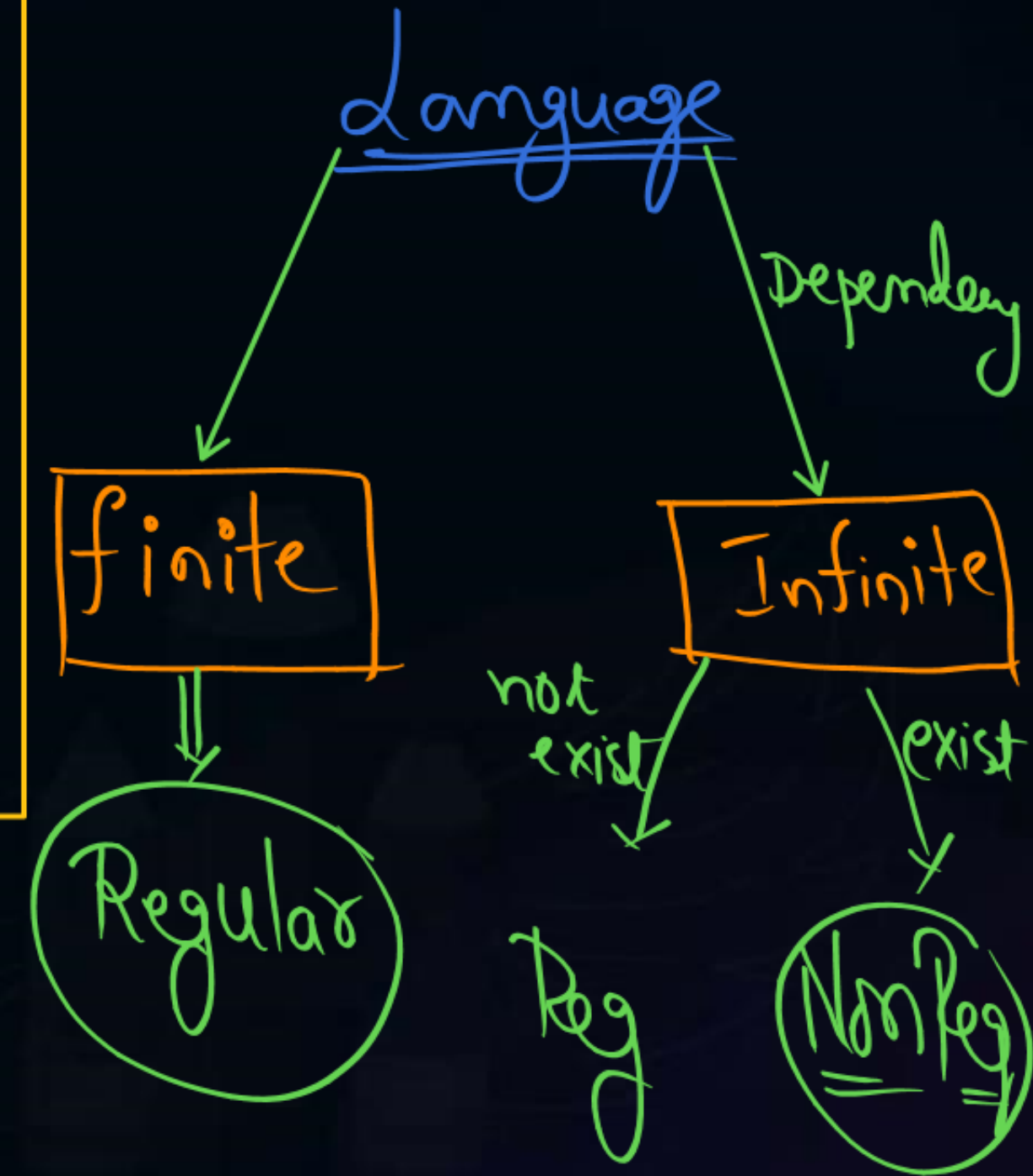
Topic : Regular Language Detection

Which of these languages are Regular?

- ① $L = \{a^n b^n c^n \mid 1 \leq n \leq 1000\}$ $\rightarrow$ finite language $\rightarrow$ Regular
- ② $L = \{a^n b^m \mid n + m = 10\}$ $\rightarrow$ finite $\rightarrow$ Regular
- ③ $L = \{a^n b^m \mid n - m = 5\}$ $\rightarrow \{a^{m+5} b^m\}$ $\rightarrow$ Infinite language Dependency Non Regular
 $n = m + 5$
- ④ $L = \{a^n b^m \mid n * m = 100\}$ $\rightarrow$ finite $\Rightarrow$ Regular
- ⑤ $L = \{a^n b^m \mid n = 2m + 1\}$ $\rightarrow \{a^{2m+1} b^m\}$ $\Rightarrow$ Non Regular
- ⑥ $L = \{a^n b^m \mid n > m\}$ $\Rightarrow$ Non Regular
- ⑦ $L = \{a^n b^m \mid (n > m \text{ and } n < m)\}$ $\Rightarrow$ Regular
 $\{ \} \Rightarrow \emptyset$

$\{a^n b^m\}$ Regular Language

finite language
 $\{a^n b^n \mid n \leq 10\}$





Topic : Regular Language Detection

- ① Every finite language is regular.
- ② Every regular language need not be finite.
- All palindrome languages over more than one symbol are non regular.
- Any infinite language require infinite memory then it is non regular.

$$L_1 = \{a^n b^m / n \geq m \text{ (and) } n \leq m\} = \{a^n b^n\} \rightarrow \text{Non Regular}$$



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- ⑧ $L = \{a^n b^m \mid n > m \text{ (or) } n < m\} = \{a^n b^m \mid n \neq m\} \rightarrow \text{Non Regular}$
- ⑨ $L = \{a^n b^m c^{n+m} \mid n, m \geq 1\} \rightarrow \text{Non Regular}$
- ⑩ $L = \{a^n b^{n+m} \mid n, m \geq 1\} \Rightarrow \text{Non Reg}$
- ⑪ $L = \{a^n b^{2m} c^{3k} \mid n, m, k \geq 0\} \Rightarrow \text{Regular}$
- ⑫ $L = \{a^n b^{m^2} c^{k^3} \mid n, m \geq 1\} \Rightarrow \text{Non Regular}$
- ⑬ $L = \{a^{2^n} \mid n \geq 0\} \Rightarrow \text{Non Regular}$
- ⑭ $L = \{a^{n^2} \mid n \geq 1\} \Rightarrow \text{Non Regular}$



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⑮ $L = \{a^{n^n} \mid n \geq 1\} \Rightarrow \text{Non Regular}$

⑯ $L = \{a^{100^{100^{100}}} \mid \} \Rightarrow \underline{\text{Regular}}$



~~⑰~~ $L = \{(a^p)^* \mid p \text{ is prime number}\} \xRightarrow{\text{Regular}} \{(a^2)^* (a^3)^* (a^5)^* (a^7)^* (a^{11})^* \dots\} = \{\epsilon, a^2, a^3, a^4, a^5, a^6, a^7, \dots\}$

⑱ $L = \{a^p \mid p \text{ is prime number}\} \Rightarrow \underline{\text{Non Regular}}$

⑲ $L = \{a^k \mid k \text{ is even number}\} \Rightarrow \text{Regular}$

$$L = \{ww^R \mid w \in \{a, b\}^*\}$$

(Q) Which of the following is Non Regular?

(a) $L = \{(a^n)^* \mid n \geq 1\} = \{(a^1)^* \dots\} = a^*$

(b) $L = \{(a^{2^n})^* \mid n \geq 0\} = \{(a^1)^* \dots\} = a^*$

(c) $L = \{(a^{n!})^* \mid n \geq 1\} = \{(a^1)^* \dots\} = a^*$

~~(d)~~ None



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Home Work

$$(29) L = \{wbw^R \mid w \in \{a\}^*\}$$

$$(30) L = \left\{ x \mid x \in \{a, b\}^* \right. \\ \left. n_a(x) \bmod 3 = n_b(x) \bmod 3 \right\}$$

$$(31) L = \left\{ x \mid x \in \{a, b\}^* \right. \\ \left. n_a(x) \bmod 3 > n_b(x) \bmod 3 \right\}$$

$$(32) L = \left\{ x \mid x \in \{a, b\}^* \right. \\ \left. n_a(x) \bmod 3 \neq n_b(x) \bmod 3 \right\}$$

$$(33) L = \left\{ x \mid x \in \{a, b, c\}^* \right. \\ \left. n_a(x) \neq n_b(x) \right\}$$



Topic : Regular Language Detection

$((()))(())$ - - - -

- (20) Set of all balanced parentheses $\Rightarrow$ Non Regular

$)) (($

- (21) Equal number of open and close parentheses } Non Regular

- (22) Even length palindrome strings of English Language. $\Rightarrow$ Non Reg

- (23) Odd length palindrome strings of Hindi Language $\rightarrow$ Non Regular

$\{\epsilon, abba, abbbba, \dots\}$

(24) $L = \{ \underline{W} \underline{W}^R \mid W \in (a+b)^* \} \Rightarrow \underline{\text{Non Regular}}$

$\overset{W}{\text{ma}} \overset{W^R}{\text{da m}}$

(25) $L = \{ \overset{ababa}{\underline{W} \underline{X} \underline{W}^R} \mid W, X \in (a+b)^* \} \Rightarrow (a+b)^* \} \underline{\text{Regular}}$

$\{ \underline{(a+b)^*} \} + \underline{ababa} +$

(26) $L = \{ \underline{W} \underline{C} \underline{W}^R \mid W \in (a+b)^* \} \Rightarrow \underline{\text{Non Regular}}$

(27) $L = \{ \overset{\epsilon, aa, abab}{\underline{W} \underline{W}} \mid W \in (a+b)^* \} \Rightarrow \underline{\text{Non Regular}}$

(28) $L = \{ \underline{W} \underline{C} \underline{W} \mid W \in (a+b)^* \} \Rightarrow \underline{\text{Non Regular}}$



THANK - YOU